

PHASE 2: REQUIREMENT ANALYSIS

1. Introduction

LegalEase is an AI-powered legal document generation application designed to simplify the creation of structured legal documents. It accepts user inputs such as document type, parties involved, terms and conditions, and effective date, and uses AI to generate the required document.

2. User Requirements

The system should allow users to:

- Select or enter the document type.
- Enter the parties involved.
- Provide terms and conditions.
- Enter the effective date.
- Generate a legal document.
- Preview and edit the generated document.
- Download the document in TXT, DOCX, and PDF formats.

3. Functional Requirements

3.1 User Input

The application should provide fields for document type, parties, terms, and effective date.

3.2 AI Document Generation

The system should process the user's inputs and generate structured legal content using the Gemini AI model.

3.3 Preview and Editing

The generated document should be displayed for preview and allow users to edit the content.

3.4 Document Download

Users should be able to download the generated document in TXT, DOCX, or PDF format.

The project document specifies these input, generation, editing, preview, and export functionalities.

4. Non-Functional Requirements

4.1 Usability

The application should provide a simple and user-friendly interface so that users can enter information and generate documents easily.

4.2 Performance

The system should process the user's input and provide the generated document efficiently.

4.3 Security

Sensitive user information, including the Gemini API key, should be stored securely using environment variables such as a .env file.

4.4 Maintainability

The application should use a modular structure with separate frontend, backend, and AI integration components.

5. Software Requirements

- Python 3.10 or above
- FastAPI
- Uvicorn
- Streamlit
- Google Generative AI SDK
- python-docx
- FPDF
- Pillow

- Requests
- python-dotenv

The project document specifically lists Python 3.10+, FastAPI, Streamlit, Google Generative AI SDK, and the supporting libraries as development requirements.

6. System Requirements

A computer capable of running Python, the required libraries, FastAPI backend, and Streamlit frontend is required for local development and execution.